ASSESSMENT TOOL – 1

Course Code	CSA0702	**Course Title**	Computer Networks
CO Assessed	CO2	**Assessment**	Analytical Problem Solving
Weightage		**Total Marks**	

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Section / Batch	ECE-2024	**Date of Submission**	29/7/26
Faculty Incharge	Dr R S Selvan	**Assessment**	Individual Submission

Total Marks	20	**Obtained Marks**	20

Code of Conduct:

I Cindy R/192412571 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

University Network Solving

Given:

Network : 192.168.20.0/24

Lab A → /26

Lab B → /27

Lab C → /28

LAB A (/26)

Subnet Mask : 255.255.255.192

Network Address : 192.168.20.0

Host Range : 192.168.20.1 – 192.168.20.62

Broadcast Address : 192.168.20.63

Usable Hosts :

$2^6 - 2 = 62$

LAB B (/27)

Subnet Mask : 255.255.2555.224

Network Address : 192.168.20.64

Hosted Range : 192.168.20.65 – 192.168.20.94

Broadcast Address : 192.168.20.95

Usable Host :

$2^5 - 2 = 30$

LAB C (/28)

Subnet Mask : 255 285.255.240

Network Address : 192.168.20.96

Host Range: 192.168.20.97 - 192.168.20.110

Broadcast Address: 192.168.20.111

Usable Hosts:

$$2^4 - 2 = 14$$

2. Analyze the Given systems

Given:

System 1

IP Addresses: 192.168.1.130

Mask: 255.255.255.192 (/26)

System 2

IP Address: 192.168.1.190

Mask: 255.255.255.192 (/26)

Step 1: Subnet Range

A /26 subnet has a block size of 64

Subnets:

192.168.1.0-63

192.168.1.64-127

192.168.1.128-191

192.168.1.192-255

Both IP addresses fall within: 192.168.1.128/26.

Network Information:

Network address: 192.168.1.128

Broadcast address: 192.168.1.191

t Range: 192.168.1.129 - 192.168.1.190

erification:

system1 - 192.168.1.180 - Valid ✓

system2 - 192.168.1.190 - Valid ✓

Both belong to the same subnet

Usable Host

$$2^6 - 2 = 62$$

3. Organization Network Allocation

Given:

Network: 172.16.0.0/16

Departments: HR → /25

Sales → /26

IT → /27

Admin → /28

HR (/25)

Network: 172.16.0.0

Host Range: 172.16.0.1 → 172.16.0.126

Broadcast: 172.16.0.127

Hosts: 126

Sales (/26)

Network: 172.16.0.128

Host Range: 172.16.0.129 - 172.16.0.190

Broadcast: 172.16.0.191

Hosts : 62

IT (/27)

Network : 172.16.0.192

Host Range : 172.16.0.193 – 172.16.0.222

Broadcast : 172.16.0.223

Hosts : 30

Admin (/28)

Network : 172.16.0.224

Host Range : 172.16.0.225 – 172.16.0.238

Broadcast : 172.16.0.239

Hosts : 14

Remaining Unused Range

172.16.0.240 – 172.16.255.255

4. Subnet Identification:

Given:

Network : 172.16.10.0/24

Subnet Mask : /27

Given IP : 172.16.10.78

Step 1: No. of Subnets

Original Network : /24

New Network : /27

Borrowed Bits : 3

No. of subnets : $2^3 = 8$, Total subnets = 8.

p2: Host per subnet

Remaining Host Bits : 5

$2^5 - 2 = 30$

Usable Hosts = 30

step3: subnet of 172.16.10.78

Block size: 32

subnets:

* 0-31
* 32-63
* 64-95
* 96-127
* 128-159
* 160-191
* 192-223
* 224-255

172.16.10.78 belongs to 172.16.10.64/27

Network Address: 172.16.10.64

Broadcast Address: 172.16.10.95

Host Range: 172.16.10.65 - 172.16.10.94

Does 172.16.10.95 belong to the same subnet?

Yes. However, 172.16.10.95 is the broadcast address, so it cannot be assigned to a host.

5. IPv6 Addressing

Given Address

2001:0dB8:0000:0000:0000:ff00:0042:8329

Compressed form: 2001:db8::ff00:42:8329

Expanded form:

2001:0db8:0000:0000:0000:ff00:0042:8329

Network prefix (/64)

2001:db8:0000:0000::/64

Host bits:

$128 - 64 = 64$

Total Host Addresses $= 2^{64} = 18,446,744,073,709,551,616$